COMP9517 – Computer Vision – Individual Project

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***Abstract*—This report describes the methodologies adopted to classify images into those containing pedestrians and those without pedestrians. Pixel intensity values of grayscale image and Histogram of Oriented Gradients (HOG) methods were used as feature extraction techniques. The resultant features were classified using supervised learning techniques namely Support Vector Machine (SVM) and K-Nearest Neighbour (KNN). Appropriate subset of the provided training data was used to perform prediction on the provided full test dataset. The best results were obtained by using HOG followed by SVM.**

***Keywords*—HOG, SVM, KNN**

I. Introduction and Background

The task is to classify images into those containing pedestrians and those without pedestrains using non-deep learning methods for feature extraction. Supervised or unsupervised classification technique can be used for this task. Also the task specification gives us the freedom to choose the entire training dataset or a subset of it.

The data provided for train and test were present across multiple directories. The training data is of size 72,114 with 22,114 positive instances and 50,000 negative instances. The test data is of size 9,457 with 3,457 positive instances and 6,000 negative instances. Each instance is an image of width 64 pixels and height 80 pixels (64 x 80)

Histogram of Oriented Gradients (HOG) as mentioned in [1] is found to be a suitable feature extraction technique for this task when used in conjuction with Support Vector Machines (SVM) as a classification model.

II. Method

In total, 8 combination of techniques were used for this classification task. The combination of techniques involved a technique for feature extraction, a technique for classification and a subset of data. For classification, supervised learning was used. The details are as given below :

A. Feature Extraction

To test the baseline performance, the simplest possible feature namely all pixel values of grayscale image was used. To get better results a better feature selection method namely HOG which is shown to perform well for pedestrian detection [1] was used. HOG feature selection was used with 2 different values for cells per block – 2x2 cells per block and 1x1 cell per block.

B. Models

K-Nearest Neighbour (KNN) was used since it is a simple model for baseline performance measurement. To obtain improved results, SVM was used, which is known to perform well for this task as mentioned in [1]

III. Experiment

As first step, the training and testing data provided across multiple directories were extracted. Various combinations of feature extraction and classification models on various subsets of data were tried followed by evaluation metrics.

Accuracy and AUC were used as evaluation metrics. Accuracy is the ratio of number of samples correctly classified out of the total samples. AUC or Area Under the Receiver Operating Characteristics Curve is the area under the plot of True Positive Rate Vs False Positive Rate. For both the metrics, a value close to 1 signifies a good model. For an imbalanced class prediction problem, as in this task, a naive model which predicts the majority class will obtain a high accuracy. But the AUC value for such a predictor will be quite low. So reporting both accuracy and AUC can give a good estimate of the reliability of the model. In addition to these metrics, for the best model, the confusion matrix has also been shown. This gives us the count of True Positives (TP), True Negatives (TN), False Positives (FP) and False Negatives (FN).

To obtain a baseline performance, a model was created taking grayscale pixel intensity values as features. This gave 5120 (80x64) features. A random subset of size 8,000 from the training data and a random subset of size 2,000 from the test data was used for this. This was run through both SVM and KNN. SVM had an accuracy of 0.913 and an AUC of 0.973. KNN performed poorly with an accuracy of 0.658 and an AUC of 0.639.

Next, the same subset of images was used with HOG for feature extraction. After resizing the input colour image to size 64x128, HOG was applied with 2x2 cells per block. This gave 3780 features per image. Running SVM on this feature set gave an accuracy of 0.990 and AUC of 0.999. KNN on the same feature set gave accuracy of 0.808 and AUC of 0.843 . Though the results were good, the training time was quite high.

With the idea of obtaining fewer features per image, HOG with 1x1 cell per block was used next for feature extraction.

HOG with 1x1 cell per block gave 1152 features per image. There were visibly no difference in the HOG feature descriptors for 2x2 cells per block and 1x1 cells per block. So the understanding was that the final evaluation metric values would be almost same for both, which was indeed the case. The training time for both SVM and KNN, on this feature set was half as compared to the previous HOG feature set. In the same data subset as previously mentioned, but with the new feature set, SVM gave an accuracy of 0.984 and AUC of 0.998. KNN reported an accuracy of 0.744 and AUC of 0.774.

Given that the results are almost same for both, and since the execution time for HOG with 1x1 cells per block was lower, it was decided that this method would be used for feature extraction for training on the entire test set. Also, since SVM consistently provided better results than KNN, only SVM was used while training with larger subsets of data.

As the next method - data subset combination, HOG with 1x1 cell per block feature extraction method was applied on the complete training and test data set. This followed by SVM for classification and gave a remarkable accuracy of 0.991 and AUC of 0.999. This took about 25 minutes on my local machine for training.

As next step, the objective was to try and use a smaller data subset, so as to obtain quicker training time with no significant drop in accuracy and AUC. So instead of the full dataset, subset of training data was chosen so that it is 4 times that of full test set size, giving train : test ratio of 80:20. i.e. random subset of size 37828 for training and the full test data of size 9457. Similar to previous case, HOG with 1 cell per block for feature extraction and SVM for prediction was used on this dataset. This gave an accuracy of 0.989 and AUC of 0.999. These values are almost same as that on the complete training data, but the execution time was 8.65 minutes which is almost one third of the training time on full training dataset. For this reason, this is being taken as the best model.

IV. RESULTS AND DISCUSSION

The results of various feature extraction – model and data subset combination are as specified in the table below.

Table 1. Results

Data Size	**Feature Extraction**	**Classification Model**	**Accuracy**	**AUC**
Training : 8000 Test : 2000	Grayscale pixel intensities (5120 features)	KNN	0.658	0.639
		SVM	0.913	0.973
	HOG with 2x2 cells per block (3780 features)	KNN	0.808	0.843
		SVM	0.990	0.999
	HOG with 1x1 per block (1152 features)	KNN	0.744	0.774
		SVM	0.984	0.998
Training : 72114 (full data) Test : 9457 (full data)	HOG with 1x1 per block (1152 features)	SVM	0.991	0.999
Training : 37828 Test : 9457 (full data)	HOG with 1x1 per block (1152 features)	SVM	0.989	0.999

The best model is taken to be the last entry in the above table, which trains on 37828 instances and uses full test data extracting features from a resized image of 64x128 using HoG with 1 cell per block, and then using SVM for classification giving **accuracy of 0.989 and AUC of 0.999**

The confusion matrix for the above mentioned best result is given below :

Table 2. Confusion Matrix

	Predicted = 0	Predicted = 1
Actual = 0	5953	47
Actual = 1	59	3398

In the above table, value of 0 denotes prediction as not containing pedestrian and the other for value of 1.

As seen in the above table for the prediction in the test set, TP = 5953, TN = 3398, FN = 47, FP = 59

Given below are some images at various stages of feature extraction :

Fig. 1.1. Positive sample image : 64 x 80

Fig. 1.2. Resized positive sample : 64 x 128

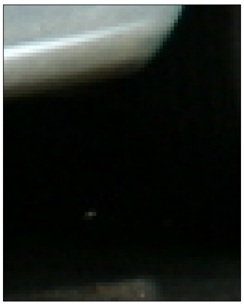

Fig. 2.1. Negative sample image : 64 x 80

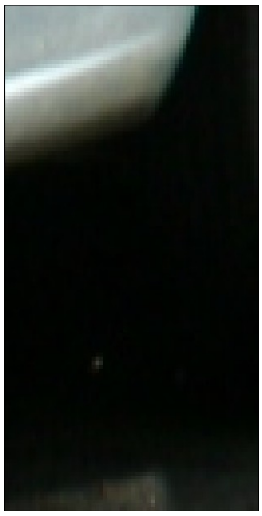

Fig. 2.2. Resized negative sample : 64 x 128

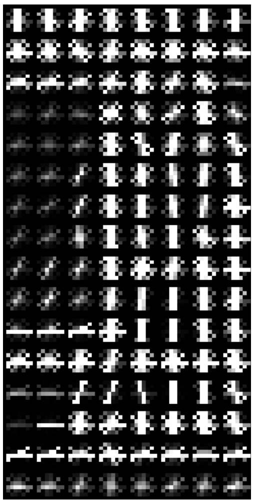

Fig. 1.3. HOG feature descriptor : 2x2 cells per block

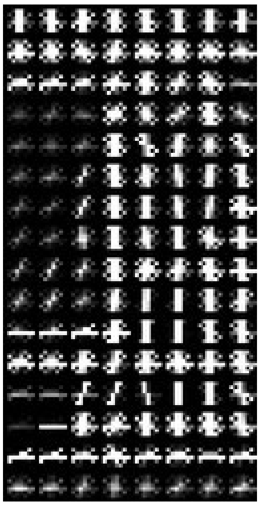

Fig. 1.4. HOG feature descriptor : 1x1 cell per block

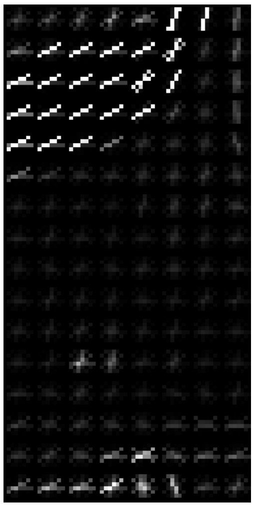

Fig. 2.3. HOG feature descriptor : 2x2 cells per block

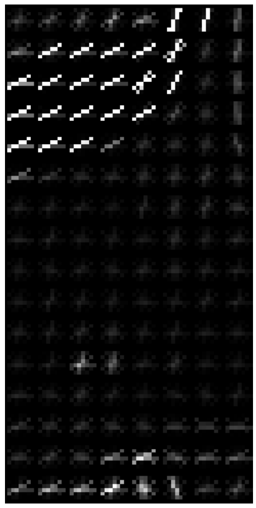

Fig. 2.4. HOG feature descriptor : 1x1 cell per block

References

[1] *N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," Computer Vision and Pattern Recognition 2005. https://doi.org/10.1109/CVPR.2005.177*

[2] https://www.analyticsvidhya.com/blog/2019/09/feature-engineering-images-introduction-hog-feature-descriptor/

[3] https://scikit-image.org/docs/dev/api/skimage.feature.html#skimage.feature.hog